0580_w20_ms_43_Q7

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(a)(i) | $(8 - x)(3 + x)$ | **2** | **M1** for $8(3 + x) - x(3 + x)$<br>or $3(8 - x) + x(8 - x)$<br>or $(a - x)(b + x)$ where $ab = 24$ or $a - b = 5$ |
| 7(a)(ii) | [$a$ = ] –3<br>[$b$ = ] 8<br>[c = ] 24 | **3** | **FT** *their* **(a)(i)** for $a$ and $b$<br>**B1FT** for each of $a$ and $b$ or both correct but reversed<br>**B1** for [$c$ =] 24 |
| 7(a)(iii) | 8 | **3** | **M2** for $5 - 2x$<br>or **M1** for $-2x$ or $5 - kx$, $k \neq 0$ |
| 7(b)(i) | Correct sketch:<br>positive cubic shape and max on the $y$-axis or to the right of $y$-axis<br>with one root at (–1, 0)<br>and<br>turning point at (3, 0)<br>and<br>$y$-intercept at (0, 9) all labelled | **4** | **B1** for positive cubic shape with max on the $y$-axis or to the right of $y$-axis<br>**B1** for root at (–1, 0)<br>**B1** for turning point at (3, 0)<br>**B1** for $y$-intercept (0, 9)<br><br>If 0 score **SC1** for all three intercepts on axes identified |
| 7(b)(ii) | $x^3 - 5x^2 + 3x + 9$ final answer | **3** | **B2** for correct expansion of three brackets unsimplified<br><br>**B1** for correct expansion of two brackets with at least 3 terms correct |